DDD-R-30
SEPTEMBER 20, 1956
SUPERSEDING
Int. Fed. Spec. DDD–R–0030 (NAVY–SHIPS)
March 4, 1955, and
Fed. Spec. DDD–W–415a
July 31, 1952

FEDERAL SPECIFICATION

RAGS, WIPING, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers cotton wiping rags primarily intended for use in wiping machinery and paint, and for miscellaneous cleaning.

1.2 Classification.

1.2.1 *Classes.*—Cotton wiping rags shall be of the following classes, as specified (see 6.1):

Class 1.—Unused fabrics.

Class 2.—Reclaimed fabrics (see 3.1.2).

1.2.2 *Colors.*—Wiping rags shall be furnished in the following colors, as specified (see 6.1):

Undyed.

Mixed colors (may include undyed rags).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

T–T–911—Twine; Jute.

QQ–S–790 — Strapping, Round; Steel, Bare and Zinc-Coated.

CCC–T–191—Textile Test Methods.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

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Department of Commerce Publication:

Commercial Standard CS 11–41—Moisture Regains of Cotton Yarns.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Class 1, unused fabrics.* — Class 1 wiping rags shall be of soft, absorbent cotton fabrics, free from stain, dust, abrasive material, and alkali. The wiping cloths shall consist of clean mill ends, mill remnants, or both. Terry woven fabrics, heavily napped fabrics, mesh fabrics having openings larger than ⅛-inch square, or weighing less than three ounces per square yard, fabrics woven with hard twisted yarns, and starched or stiffened fabrics will not be acceptable.

3.1.2 *Class 2, reclaimed fabrics.*—Class 2 wiping rags shall be soft, absorbent cotton fabrics consisting principally of fabrics reclaimed from household articles and garments. The following wiping rags shall be considered defective:

(*a*) Rags having stains which retard absorbency (see 3.4) or which cover more than one-tenth of the area of the rag.

(*b*) Rags containing dirt, dust, abrasives, and other clearly noticeable nonfibrous materials or an aggregate of more than one square inch of hardened surface.

(*c*) Rags made of mesh fabric having an opening larger than ⅛-inch square or made of mesh fabric weighing less than three ounces per square yard. (Cotton lace, and scrim are considered in these categories.)

(*d*) Rags so badly worn or tendered as to be unsuited for the intended use.

(*e*) Tattered parts less than 3 inches in width and 6 inches or more long. (A tatter is a fabric part protruding from the body of the cloth.) Edges frayed or raveled continuously or intermittently, to a depth of over 2 inches.

(*f*) Rags made of pants and overalls in which the crotch, cuff, cuff hems, waist bands and pockets have not been completely eliminated.

(*g*) Rags made of unopened sleeves, unopened legs and unopened tubular fabrics.

(*h*) Rags over 9 ounces per square yard having pockets, reinforcements (including collars, cuffs and buttonhole and waistband reinforcements including welting and piping, such as is used on slip covers etc.), unopened hems over 1 inch wide and patches. However, tight patches not over 4 square inches in area will be acceptable.

(*i*) Rags containing buttons, hooks, eyes, pins, or any other metallic or plastic fittings.

(*j*) Rags made of U.S. flags, flags of other nations, or remnants thereof.

3.2 Size.

3.2.1 *Class 1, unused fabrics.*—Each wiping rag shall have an area of not less than 2 square feet, and shall be not less than 9 inches wide and not more than 44 inches long. Rags which have been sewn together to obtain the minimum size will be accepted provided not more than two pieces of fabric, both seamless, have been used.

3.2.2 *Class 2, reclaimed fabrics.*—Each wiping rag made from fabrics weighing less than 5 ounces per square yard shall have an area of not less than 2 square feet, and shall be not less than 9 inches wide and not more than 44 inches long, except that it may be

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not less than 6 inches wide for not more than 6 inches in length. Each wiping rag made of fabrics weighing more than 5 ounces per square yard shall have the area and measurements specified above for rags made of fabrics weighing less than 5 ounces per square yard, except that rags with an area of from 225 to 288 square inches will be acceptable when their minimum dimension is not less than 10 inches.

3.2.2.1 The width and length dimensions specified in 3.2.2 are defined as follows: The minimum width of a rag shall be the measurement between two opposing points on the perimeter closest to each other. The maximum length shall be the distance between two points on edge of a rag most distant from each other. When rags are nearly rectangular, the long side shall be considered the maximum length.

3.3 Weight.—The wiping rags shall weigh not more than 12 ounces per square yard nor less than 2 ounces per square yard. At least 35 percent of the weight of the rags in each bale shall weigh between 2 and 5 ounces per square yard.

3.4 Absorbency.—Cotton wiping rags shall be absorbent, and when tested in accordance with 4.2.3 the rags shall be considered absorbent if the water and oil are absorbed into the fabric within 30 seconds. However, if the water or oil is not absorbed but remains wholly or partly as a film on the surface of the fabric or on the supporting surface, the rag shall be considered nonabsorbent.

3.5 Washing and sterilization.

3.5.1 *Class 1, unused fabrics.*—Class 1 wiping rags may be washed.

3.5.2 *Class 2, reclaimed fabrics.*—Class 2 wiping rags shall be washed and sterilized. (This processing shall be done within the continental limits or territories of the United States.) (See 6.1.1 (*a*)).

3.6 Moisture content.—Rags shall not have a moisture content greater than 10 percent.

3.7 Basis of purchase.—Wiping rags shall be purchased by net weight provided the moisture content as determined in 4.3 is 6.5 ± 1 percent (based on commercial moisture regain of 7 percent as covered in Commercial Standard, Moisture Regains of Cotton Yarns, CS–11–41). Deliveries containing less than 5.5 and more than 7.5 percent moisture shall be purchased by net weight adjusted to a basis of rags containing 6.5 percent moisture (see 6.1.1 (b) and 6.2).

3.8 Workmanship.—The wiping rags shall be clean, and free from objectionable odor.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of not more than 50,000 pounds of wiping rags of one class and color offered for delivery at one time.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of wiping rags shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics, with lot acceptance based on the sampling inspection requirements in accordance with Military Standard MIL–STD–105. The unit of product shall be 1 pound of rags. For normal inspection the sampling and acceptable quality level (AQL) shall be as shown in table I. Tightened and reduced inspection shall be in accordance with section 9 of Military Standard MIL–STD–105.

4.1.2.1 *Class 1 wiping rags.*—The acceptable quality level shall be 4.0 percent.

4.1.2.2 *Class 2 wiping rags.*—The acceptable quality level shall be 2.5 percent for major defects and 6.5 percent for major and minor defects combined (see 4.2).

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TABLE I.—*Sampling for visual and dimensional inspection*

Pound of rags in inspection lot	Pounds of rags in sample	Pounds of defectives acceptable[1]		
		Class 1	Class 2	
		AQL (percent) 4.0	Major defects AQL (percent) 2.5	Major and minor defects combined AQL (percent) 6.5
110 and under	15	1	0	2
111 to 180	25	2	1	3
181 to 300	35	3	2	5
301 to 500	50	4	3	6
501 to 800	75	6	4	9
801 to 1,300	110	8	6	12
1,301 to 3,200	150	11	8	17
3,201 to 8,000	225	17	11	24
8,001 to 22,000	300	20	14	32
22,001 to 50,000 incl.	450	29	20	43

[1] If the pounds of defectives exceed the amounts permitted in table I, the lot shall be rejected.

4.1.3 *Sampling for determination of moisture content.*—From each lot offered for Government inspection sample bales shall be selected at random in accordance with table II (inspection level L–3) of the appendix to Military Standard MIL–STD–105). From each sample bale three samples of 1 pound (approximately) shall be taken; two from opposite sides of the bale and the third from near the center. Each of these samples shall be kept separate so that moisture content can be determined for each sample separately. If equipment is not available in the contractor's plant for moisture determination, each sample shall be placed in an airtight container as soon as possible and forwarded to the laboratory designated by the procuring agency concerned for moisture content determination.

TABLE II.—*Sampling for moisture determination*

Lot size, number of bales	Sample, number of bales to be tested
1 to 180	1
181 to 500	2
501 to 800	3
801 to 1,000	5

4.1.4 *Sampling for quantity verification.*—From each lot a sample of bales shall be selected in accordance with table III for weighing to verify the gross weights (5.2) and to determine the amount of tare. Each of the sample bales shall be weighed separately for comparison with the contractor's markings. The average weight shall be computed and this average shall be multiplied by the number of bales to arrive at the total gross weight of the bales. The adjustment for tare shall be computed in a similar manner from the tare of the bales used in the sampling of 4.1.2. The adjustment for moisture shall be made as specified in 3.7 and 4.3.

TABLE III.—*Sampling for computation of weight*

Number of bales in inspection lot	Number of bales to be weighed
2 to 15	2
16 to 40	3
41 to 65	5
66 to 110	7
111 to 180	10
181 to 300	15
301 to 500	25
501 to 800	35
801 to 1,000	50

4.2 Visual and dimensional inspection.—The pounds of rags selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. If the total weight of nonconforming rags in the sample exceeds the applicable

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acceptance number (in pounds), the entire lot shall be rejected. The weight of class 2 nonconforming rags shall be divided into the following major and minor defect classifications:

Major defects:

(*a*) Rags not made of cotton.
(*b*) Rags having the defects specified in 3.1.2(a), (b), (c), (d), and (j).
(*c*) Rags not within the dimensions specified in 3.2.2.
(*d*) Rags not within the weight requirements of 3.3.
(*e*) Rags not having the absorbency specified in 3.4.

Minor defects:

(*a*) Rags having the defects specified in 3.1.2 (e), (f), (g), (h), and (i).
(*b*) Rags having objectionable odor.

4.2.1 *Examination for tenderness.*—Rags which are in question as to tenderness (see 3.1.2 (*d*)) shall be examined in the following manner. The rag shall be grasped on opposite edges so that it remains fully opened and flat. There shall be approximately 3 inches of material gathered in each hand. With the arms fully extended, the rag shall be subjected to a steady outward force. If the rag parts, it shall be considered tender.

4.2.2 *Examination for weight per square yard.*—In cases of dispute or indecision as to weight of rags (see 3.3), weight shall be determined in accordance with method 5041 of Federal Specification CCC–T–191. In order to allow for some error of measurement in determining the weight-per-square-yard of cloths, a tolerance of plus 10 percent on the maximum and minus 10 percent on the minimum requirement will be considered acceptable.

4.2.3 *Examination for absorbency.*

4.2.3.1 Rags which are in question as to absorbency (see 3.4) shall be examined in the following manner: The sample rag shall be laid on a nonabsorbent surface and several drops of water and a light machine oil applied. After 30 seconds the sample shall be examined and the absorption or nonabsorption of the water and oil reported.

4.2.3.2 In cases of dispute, drops 0.04 milliliter in volume shall be applied to the sample rags through a capillary by allowing the drops to fall freely into the rag from a height of ½ to 2 inches.

4.3 **Test for moisture content.**—The samples selected in accordance with 4.1.3 shall be accurately weighed and then dried in an oven at 221° to 230°F. (105° to 110°C.) until the weights from two consecutive weighings, taken not less than 10 minutes apart, differ by not more than 0.1 percent of the first of the two weights. The moisture content is the difference between the weight of the rags as sampled (*Wo*) and their "dry weight" (*Wd*), expressed as a percentage of the original (sampled) weight:

$$\text{Moisture content (percent)} = \frac{Wo - Wd}{Wo} \times 100$$

The average of the moisture contents of the individual samples shall be considered the moisture content of the entire lot represented by the samples. If the average moisture content of the individual samples exceeds 10 percent, the entire lot shall be rejected.

4.4 **Rejected lots.**—A lot which has been rejected under the provision of 4.2 or 4.3 may be resubmitted for Government inspection and tests provided the contractor, after being informed of the reasons for rejection, has inspected or tested each bale in the lot for the deficiencies noted, and has removed all nonconforming rags. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.*—The wiping rags shall be furnished in approximately 50-pound quantity compressed bales. The bales shall be compressed to a density of not more than 6½ cubic feet per hundred pounds with no dimension of the bale exceeding 20 inches. The baling material shall be clean and serviceable burlap which need not necessarily be new but must be closely woven and free from holes and markings. The material shall be of such size as to fully cover the bale with a minimum 3-inch overlap on the sides and ends.

5.1.1.1 *Bale closure.* — After compression, the baling material shall be overlapped as specified in 5.1.1, and secured with tension-tied, 16-gage, round wire steel strapping conforming to Federal Specification QQ–S–790. At the option of the supplier the wire ties may be applied either outside or underneath the baling material. Three straps will be required equally spaced between the heads or ends of the bale. The heads or ends of the bale shall be sewn with three-ply jute twine conforming to type B of Federal Specification T–T–911 or secured with steel clips or hog rings spaced at proper intervals along the seam.

5.1.2 *Level C.*—The wiping rags shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from the normal hazards of transportation.

5.1.3 For definitions and levels see 6.4.

5.2 Marking.

5.2.1 *For Military purchases.*—In addition to any special marking required in the contract or order, marking of the bales shall be as follows:

(*a*) Size, location of required markings, and marking materials (ink and tags) shall be in accordance with Military Standard MIL–STD–129.

(*b*) Identification markings shall consist of the following:

(1) Stock number
(2) Nomenclature
(3) Date packed
(4) Gross weight
(5) Cube

(*c*) Contract data marking shall consist of the following:

(1) Name and address of the contractor.
(2) Contract or order number.

(*d*) Domestic or overseas address shall be as specified in the contract or order.

5.2.2 *For nonmilitary purchases.*—Unless otherwise specified in the contract or order, each container shall be marked with the name and class of the commodity as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, the name and address of the consignee, the net weight of the wiping rags, the tare, and the gross weight.

5.2.3 *General.*—Bill of lading or shipping order number shall be marked on the container in letters not less than ¾ inch in height. Domestic address and bill of lading or shipping order number will only be required on ten percent of the shipping containers in sealed car or truck loads. The marked containers shall be placed in or adjacent to the doorways or entrance areas.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

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(*b*) Class required (see 1.2.1).
(*c*) Color required (see 1.2.2).
(*d*) Level of packing required (see 5.1).
(*e*) Non-Military marking (see 5.2.2).

6.1.1 Purchasers should specify the following contractual requirements:

(*a*) That a certified copy of any affidavit, by a responsible officer of the contractor's organization, be submitted with each shipment to the effect that the rags furnished have been washed, sterilized, and dried within the continental limits or territories of the United States (see 3.5.2).

(*b*) That net invoice weights governing payment will be on the basis of net weight of deliveries (see 3.7).

6.2 Adjustment of net weight. — Where the moisture content as determined in 4.3 is not within the limits specified in 3.7 the net weight may be adjusted by the formula:

$$W = \frac{N \times (100 - C)}{93.5}$$

where:

W = adjusted net weight.
N = net weight as delivered.
C = moisture content, in percent (4.3).

The moisture content specified in 3.7 is an approximation based on a commercial moisture regain of seven percent as defined by Commercial Standard, Moisture Regains of Cotton Yarn, CS 11–41.

6.3 Transportation description.—The transportation description appplicable to this item is:

Cloths or rags, wiping.
In machine pressed bales or machine pressed in boxes.[1]
In packages not machine pressed.
Carload weight 24,000 pounds, subject to Rule 34, Consolidated Freight Classification.
Truckload weight 24,000 pounds.

[1]Preferred method of shipment to obtain lowest transportation charge.

6.4 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q
Navy—Sh MC Or Y
Air Force.

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